

PROBABILITY *and* STATISTICS



A WORKTEXT IN GRADE 11 MATHEMATICS 4TH QUARTER

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Preface

This Worktext in Probability and Statistics tends to explain concepts and procedure as simple and clearly as possible. The problem presented as illustrated examples and activities have been carefully selected to provide the students with a thorough workout in the application of the basic principles of Probability and Statistics, the students experience the evolution necessary to the learning process.

The main objective of this worktext is to motivate students to take interest in understanding the connections between the field of reality and the field of mathematics. The topic was designed to achieve the objectives of equipping students with the fundamental knowledge on solving problems involving normal distributions.

These will serve the need of our students regardless of the course they hope to pursue in the future.

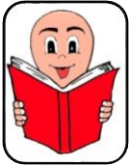
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**A WORKTEXT IN
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TOPIC: SOLVING PROBLEMS INVOLVING NORMAL DISTRIBUTIONS

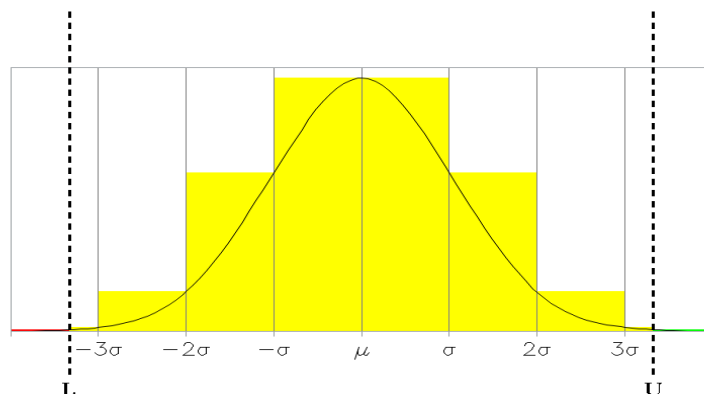


Competencies: (M11/12SP-IIIc 1-3)

- Illustrates a normal random variable and its characteristics.
- Construct a normal curve.
- Identifies regions under the normal curve corresponding to different standard normal values.

The **normal distribution** is a type of data distribution that is observed in a lot of instances in real life. There are many continuous random variables that we measure in everyday instances that have a defined normal range such as IQ scores, blood pressure, and test scores.

Normal Distribution Curve



A normal distribution can have different variations depending on its mean and standard deviation. It will be convenient to analyze only a normal distribution with a mean of 0 and a standard deviation of 1. This normal distribution is called **standard normal distribution**. By converting the possible values of a normally distributed random variable to its corresponding z-score, the normal distribution is standardized, thus making it easier to analyze.

The z-score is not entirely a different quantity. It is derived from X , which is the value that the continuous random variable assumes. To determine how the z-score is related to any value of a random variables use this formula:

$$z = \frac{X - \mu}{\sigma}$$

where:

z = z - score

X = value of the random variable

μ = population mean

σ = population standard deviation

From the equation above it is understood that each value of X has a corresponding z-score. This corresponding z-score will depend on the population mean and the standard deviation. Since a standard normal distribution has only one value for the mean and one for the standard deviation, its graph does not change. The objective is to convert the values of any normally distributed random variable to its corresponding z-score.

Example 1: A group of Grade 11 students took an entrance examination. The entrance examination score follow a normal distribution and has a mean of 75 and a standard deviation of 2.5. If a student got a score of 85, what is the corresponding z-score?

Given:

$$X = 85$$

$$\mu = 75$$

$$\sigma = 2.5$$

Solution:

$$z = \frac{X - \mu}{\sigma}$$

$$z = \frac{85 - 75}{2.5}$$

$$\underline{z = 4}$$

Example 2. Refer to problem Number 1, what is the entrance score result of a student who has a z-score of -2?

Given:

$$\mu = 75$$

$$\sigma = 2.5$$

$$z = -2$$

Solution:

$$z = \frac{X - \mu}{\sigma} \quad \text{Equate the formula to solve for } x$$

$$X = z\sigma + \mu$$

$$X = (-2)(2.5) + 75$$

$$X = -5 + 75$$

$$\underline{\underline{X = 70}}$$

To solve for the probability in a normal distribution, we cannot just shade the area bounded by the curve. To make this a lot easier, the normal distribution is converted into a standard normal distribution.



Illustrative Examples:

1. The number of tricycles that pass through Roxas Avenue everyday resembles a normal distribution with a mean of 7410 and a standard deviation of 688. On a given day, what is the probability that the number of tricycles that will pass through Roxas Avenue will be between 6000 and 7000 tricycles?



Photo owned by: Roger B. Cuyos

Solution:

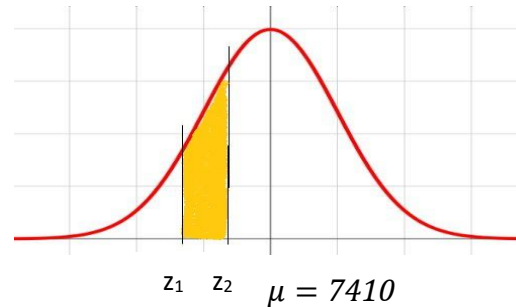
Given:

$$\mu = 7410$$

$$\sigma = 688$$

$$x_1 = 6000$$

$$x_2 = 7000$$



$$z_1 = \frac{X - \mu}{\sigma} = \frac{6000 - 7410}{688} = -2.05$$

$$z_2 = \frac{X - \mu}{\sigma} = \frac{7000 - 7410}{688} = -0.60$$

$$A_{z_1} = 0.0202$$

$$A_{z_2} = 0.2743$$

$$A_{z_1} - A_{z_2} = 0.0202 - 0.2743 = \mathbf{0.2541 \text{ or } 25.41\%}.$$

The probability of the number of tricycles that will pass through Roxas Avenue will be between 6000 and 7000 tricycles is 0.2541 or 25.41%.

2. The annual income of the family of each student in Capiz National High School resembles a normal distribution. The distribution has a mean of ₱ 96,000 with a standard deviation of ₱ 9,000. The school would like to grant scholarships to the students in the lowest 10% of the income bracket. What is the cutoff annual income for a student to be eligible for a scholarship?

Solution:

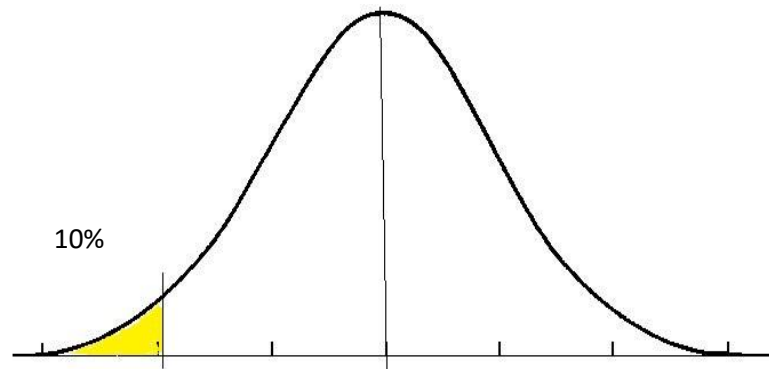
Given:

$$\mu = 96,000$$

$$\sigma = 9,000$$

$$\begin{aligned} X &= z\sigma + \mu \\ &= (-1.28)(9,000) + 96,000 \\ &= -11,520 + 96,000 \end{aligned}$$

$$\underline{X = 84,480.00}$$





Let's Try !!!

1. The Grade 11 students of Capiz National High School took a 100 item Statistics exam. The scores fit the normal distribution with a mean of 75 and a standard deviation of 10. What is the percentage of students that got a score of 80 to 85?
2. The applicants to a company took a 60 item entrance exam. Their scores resemble a normal distribution with a mean of 50 and a standard deviation of 4. If the company would like to have the top 20 percent of the applicants to go on to the next phase of the application process, what should be their cut-off score in the exam?

REFERENCES

Tizon, Melbert B. and Mesa, Helma Y., Stat Speaks, Statistics and Probability for 21st Century Learners, St. Bernadette Publishing House Corporation, 2016